

Entry Diagnostic

Ungraded readiness check | about 15 minutes | no AI or internet needed

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Purpose

This course-designed check identifies where a short refresher will help. It is not a validated aptitude test and does not reduce your grade. Write "not sure" rather than guess confidently. The instructor will point you to the bridge sections corresponding to missed items.

1. What molecule is the usual sequence reference for a genome: DNA, protein or lipid?

2. A sample has more RNA reads for an enzyme. Does this alone establish higher enzyme activity? Explain in one phrase.

3. During typical RNA splicing, which are removed from the precursor: introns or all exons?

4. How many RNA bases form a codon in the standard triplet genetic code?

5. Three donors each contribute 100 cells. How many donors provide independent donor-level replication?

6. A table has two gene rows and three sample columns. What are its dimensions?

7. What fraction of GCAA is G or C?

8. A BED interval starts at 2 and ends at 5. How many bases does it contain?

9. What is $\log_2(2)$?

10. For a Python list, is the first element at index 0 or index 1?

11. A notebook says a variable is not defined. What is one sensible first check?

12. May a patient genome with the name removed automatically be uploaded to a public chatbot?
